

Employee Constructor Demonstration

1. Objective

To develop a Java program using a class and objects to demonstrate constructors. The program uses a default constructor, a parameterized constructor with name and ID, and a parameterized constructor with complete employee details, and displays the employee information.

2. Task Description

Create an Employee class containing employee name, ID, designation, salary, and promotion status. Implement three constructors with different parameter lists and create three Employee objects in the main method. Finally, display the details of all three employees.

3. Step-wise Execution

1. Define the Employee class and declare its data members.
2. Create a default constructor and assign default values.
3. Create a constructor that accepts employee name and ID.
4. Create a constructor that accepts all employee details.
5. Define the display() method to print employee information.
6. Create object e1 using the default constructor.
7. Create object e2 using the name-and-ID constructor.
8. Create object e3 using the complete-details constructor.
9. Call display() for e1, e2, and e3.
10. Observe the employee details printed on the console.

4. Algorithm

Step 1: Start.

Step 2: Define the Employee class with name, id, designation, salary, and promotionStatus.

Step 3: Define three overloaded constructors.

Step 4: Define display() to print all employee details.

Step 5: In main(), create e1 using the default constructor.

Step 6: Create e2 using the constructor with name and ID.

Step 7: Create e3 using the constructor with all employee details.

Step 8: Call display() for each object.

Step 9: Stop.

5. UML Diagram

Employee	
Data Members	Methods / Constructors
name : String	Employee()

id : int	Employee(name, id)
designation : String	Employee(name, id, designation, salary, promotionStatus)
salary : double	display()
promotionStatus : String	

6. Program Code

```
class Employee {
    String name;
    int id;
    String designation;
    double salary;
    String promotionStatus;

    // Constructor 1: Default constructor
    Employee() {
        name = "Unknown";
        id = 0;
        designation = "Not Assigned";
        salary = 0.0;
        promotionStatus = "Not Eligible";
    }

    // Constructor 2: Constructor with name and id
    Employee(String name, int id) {
        this.name = name;
        this.id = id;
        designation = "Employee";
        salary = 20000;
        promotionStatus = "Pending";
    }

    // Constructor 3: Constructor with all employee details
    Employee(String name, int id, String designation,
        double salary, String promotionStatus) {
        this.name = name;
        this.id = id;
        this.designation = designation;
        this.salary = salary;
        this.promotionStatus = promotionStatus;
    }

    // Method to display employee details
    void display() {
        System.out.println("Name           : " + name);
        System.out.println("ID           : " + id);
        System.out.println("Designation      : " + designation);
        System.out.println("Salary          : " + salary);
        System.out.println("Promotion Status : " + promotionStatus);
        System.out.println("-----");
    }
}

public class EmployeeDemo {
    public static void main(String[] args) {

        // Using default constructor
        Employee e1 = new Employee();

        // Using constructor with name and id
        Employee e2 = new Employee("Rahul", 101);

        // Using constructor with all details
        Employee e3 = new Employee(
            "Priya", 102, "Manager", 65000, "Eligible"
        );

        e1.display();
        e2.display();
        e3.display();
    }
}
```

7. Output Screen

```
Name          : Unknown
ID            : 0
Designation   : Not Assigned
Salary       : 0.0
Promotion Status : Not Eligible
-----
```

```
Name          : Rahul
ID            : 101
Designation   : Employee
Salary       : 20000.0
Promotion Status : Pending
-----
```

```
Name          : Priya
ID            : 102
Designation   : Manager
Salary       : 65000.0
Promotion Status : Eligible
-----
```

8. Result

The Java program was executed successfully. It demonstrates constructor overloading by creating Employee objects using three different constructors and displays their details correctly.